

1 Student-t MAP Objective Derivation

1.1 Input and Assumption Nodes

Input nodes: `likelihood.student_t` and `prior.gaussian_beta`.

1.2 Derivation

The Student-t MAP approximation minimizes

$$-\sum_{i=1}^n \log t_{\nu}(y_i - x_i^{\top} \beta; 0, \sigma) + \frac{1}{2\tau^2} \|\beta\|_2^2.$$

The Student-t likelihood is not conjugate to the Gaussian prior in this workflow, so the result is an optimization-based approximation node rather than a closed-form posterior node.

1.3 Result

The result node is `approx.student_t_map`.